Ali Elouafiq
Principal Research Engineer
SQLI Digital Laboratory
2–10 rue Thierry Le Luron
92300 Levallois-Perret, France
ali@sqli.com

9 June 2026

Editor-in-Chief
The American Mathematical Monthly
Mathematical Association of America

Dear Editor,

I submit for your consideration the manuscript:

The continuous-coefficient Jensen equation:
A note on vestigial regularity hypotheses

for publication in *The American Mathematical Monthly* as an **Article**.

What the paper does. The paper shows that the continuous-coefficient Jensen equation $G(pu_1 + (1-p)u_2) = pG(u_1) + (1-p)G(u_2)$ for all $u_1, u_2 \in [0, M]$ and all $p \in [0, 1]$ forces G to be affine with *no regularity hypothesis*: a one-line endpoint substitution ($u_1{:=}M$, $u_2{:=}0$, $p{:=}v/M$) reads $G(v)$ directly off the straight line through the endpoints. The paper accompanies this one-line observation with (i) a dictionary comparing the result with the well-known case of the discrete-coefficient Jensen equation (where Hamel's pathology is real and regularity *is* needed), (ii) a structural explanation of why the two cases differ, and (iii) three application areas — expected-utility theory, the axiomatic characterization of Shannon entropy, and surrogate-loss calibration — where the same vestigial hypotheses recur.

Why the Monthly. The result is elementary enough for a second-year analysis student to verify in an afternoon, yet it is consistently missed in practice: the functional equations literature treats the two forms of Jensen's equation in adjacent sections without flagging the asymmetry, and modern applied work (utility theory, calibration, entropy) carries defensive regularity assumptions that are unnecessary precisely in the continuous-coefficient case. The Monthly is the natural home for an expository note that makes this asymmetry explicit, provides the dictionary, and invites readers to audit their own work.

Novelty and relationship to prior art. The substitution technique that closes Theorem 1 is folklore: Aczél's 1966 monograph (§2.1.4) contains a dyadic precursor proved under a continuity hypothesis, and the genre is what Kuczma's 2009 introduction describes as the part of Jensen-equation theory that does not require regularity assumptions on the function. The precise continuous-coefficient regularity-free statement of Theorem 1 does not, however, appear in this exact form in the standard references the author consulted. The contribution of this note is therefore not Theorem 1 itself but its *explicit packaging*: the dictionary contrasting the two Jensen forms, the structural explanation of the Hamel-pathology mechanism, and the identification of a recurrence pattern in derivations that produce the continuous-coefficient form as a saturated identity. A closing note records that the observation reached the author through a formalization project on a separate topic — a route that may interest readers curious about how mechanized proof can pressure-test classical heuristics.

Submission details.

- **Manuscript type:** Article (expository).
- **Page count:** 9 pages anonymous body, 10 pages with title page.
- **Word count (approximate):** 3,400 words (TeX-counted body).
- **MSC 2020 primary:** 39B22 (Functional equations for real functions).
- **MSC 2020 secondary:** 26B25 (Convexity), 91B16 (Utility theory), 94A17 (Measures of information; entropy).
- **Double-blind:** Yes. The anonymous body (`manuscript-anon.pdf`) contains no author-identifying information; the file with author details (`manuscript.pdf`) carries the title page on page 1.
- **Preprint status:** The manuscript has not been deposited on arXiv. It is not under consideration elsewhere.
- **Competing interests:** None.
- **Funding:** This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.
- **Use of generative AI:** Disclosed in the manuscript's Acknowledgments section. Anthropic's Claude Opus 4.8 was used for adversarial review of drafts and LaTeX formatting; all mathematical content was conceived, verified, and refined by the author.

I look forward to the referees' comments.

Sincerely,

Ali Elouafiq